

SECTION A (25 MARKS)

NO	ANSWER SCHEME	MARKS
1(a)	$\lim_{x \rightarrow +\infty} \sqrt{\frac{25x^2 - 4}{4x^2 + 81}} = \sqrt{\lim_{x \rightarrow +\infty} \frac{25x^2 - 4}{4x^2 + 81}}$ $= \sqrt{\lim_{x \rightarrow +\infty} \frac{\frac{25x^2 - 4}{x^2}}{\frac{4x^2 + 81}{x^2}}}$ $= \sqrt{\lim_{x \rightarrow +\infty} \frac{25 - \frac{4}{x^2}}{4 + \frac{81}{x^2}}}$ $= \sqrt{\frac{25 - 0}{4 + 0}}$ $= \frac{5}{2}$	K1 - Divide by x^2 K1 – Simplify K1 J1
1(b)	$\lim_{x \rightarrow a} \frac{\frac{1}{x} - \frac{1}{a}}{x - a} = \lim_{x \rightarrow a} \frac{\frac{a - x}{ax}}{x - a}$ $= \lim_{x \rightarrow a} \frac{a - x}{ax(x - a)}$ $= \lim_{x \rightarrow a} \frac{-(x - a)}{ax(x - a)}$ $= \lim_{x \rightarrow a} \frac{-1}{ax}$ $= \frac{-1}{a(a)}$ $= -\frac{1}{a^2}$	K1 - Equate denominator K1 - Simplify K1 Substitute J1 Answer
TOTAL		8 marks

NO	ANSWER SCHEME	MARKS
2	$3x^2 + y^2 = 2$ $6x + 2y \frac{dy}{dx} = 0$ $\frac{dy}{dx} = -\frac{3x}{y}$ $u = -3x \quad v = y$ $u' = -3 \quad v' = \frac{dy}{dx}$ $\frac{d^2y}{dx^2} = \frac{y(-3) - (-3x)\frac{dy}{dx}}{y^2}$ $= \frac{-3y + 3x\frac{dy}{dx}}{y^2}$ $= \frac{-3y + 3x\left(\frac{-3x}{y}\right)}{y^2}$ $= \frac{-3y^2 - 9x^2}{y^2}$ $= \frac{y}{y^2}$ $= \frac{-3(y^2 + 3x^2)}{y^3}$ $= \frac{-3(2)}{y^3}$ $\frac{d^2y}{dx^2} = -\frac{6}{y^3}$	K1 - Differentiate J1 - Answer K1 - Correct u and v J1 - Find u' and v' K1 - Correct formula K1 - Simplify J1 - Answer K1 - subs $3x^2 + y^2 = 2$ J1 - Answer
TOTAL		8 marks
3	$\frac{dy}{dx} = \frac{1}{3}(x^2 - 4)^{-\frac{2}{3}}(2x)$ $= \frac{2x}{3(x^2 - 4)^{\frac{2}{3}}}$ When $\frac{dy}{dx} = 0$, $\frac{2x}{3(x^2 - 4)^{\frac{2}{3}}} = 0$ $2x = 0$ $x = 0$	K1 - Differentiate J1 - Answer K1 - Find x J1 - Answer

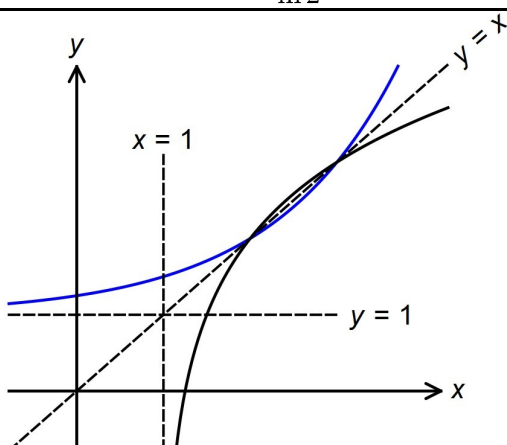
NO	ANSWER SCHEME	MARKS																				
3	$\frac{dy}{dx}$ does not exist when $3\left(x^2 - 4\right)^{\frac{2}{3}} = 0$ $x^2 - 4 = 0$ $(x + 2)(x - 2) = 0$ $x = -2, x = 2$ <table><tr><td>Interval</td><td>$(-\infty, -2)$</td><td>$(-2, 0)$</td><td>$(0, 2)$</td><td>$(2, \infty)$</td></tr><tr><td>Test value</td><td>-3</td><td>-1</td><td>1</td><td>3</td></tr><tr><td>Sign of f'</td><td>-</td><td>-</td><td>+</td><td>+</td></tr><tr><td>Conclusion</td><td>Decre.</td><td>Decre.</td><td>Incre.</td><td>Incre.</td></tr></table> Increasing intervals are $(0, 2)$ and $(2, \infty)$. Decreasing intervals are $(-\infty, -2)$ and $(-2, 0)$.	Interval	$(-\infty, -2)$	$(-2, 0)$	$(0, 2)$	$(2, \infty)$	Test value	-3	-1	1	3	Sign of f'	-	-	+	+	Conclusion	Decre.	Decre.	Incre.	Incre.	J1 - Answer K1 - Table J1 - Answer J1 - Answer
Interval	$(-\infty, -2)$	$(-2, 0)$	$(0, 2)$	$(2, \infty)$																		
Test value	-3	-1	1	3																		
Sign of f'	-	-	+	+																		
Conclusion	Decre.	Decre.	Incre.	Incre.																		
TOTAL		8 marks																				

SECTION B (75 MARKS)

NO	ANSWER SCHEME	MARKS
1	Let $z = a + bi$, $\bar{z} = a - bi$, $\bar{z} = \sqrt{21 + 20i}$ $a - bi = \sqrt{21 + 20i}$ $(a - bi)^2 = 21 + 20i$ $a^2 - 2abi + b^2i^2 = 21 + 20i$ $a^2 - b^2 - 2abi = 21 + 20i$ Compare: $a^2 - b^2 = 21 \quad -2ab = 20$ $b = -\frac{10}{a}$ $a^2 - \left(-\frac{10}{a}\right)^2 = 21$ $a^2 - \frac{100}{a^2} = 21$ $\frac{a^4 - 100}{a^2} = 21$ $a^4 - 21a^2 - 100 = 0$ $(a^2 + 4)(a^2 - 25) = 0$ $a^2 = -4 \quad a^2 = 25$ $a = \pm 5$ Since the real part is negative, thus $a = -5, \quad b = -\frac{10}{-5} = 2$ $\therefore z = -5 + 2i$	K1 - Substitute $\bar{z} = a - bi$ and squaring both sides K1 - Expand and compare K1 - Try to solve J1 - Choose a J1 - Answer
TOTAL		5 marks
2(a)	$x^6 e^{-4 \ln x} = 2x + 15$ $x^6 e^{\ln x^{-4}} = 2x + 15$ $x^6 x^{-4} = 2x + 15$ $x^2 = 2x + 15$ $x^2 - 2x - 15 = 0$ $(x - 5)(x + 3) = 0$ $x = 5, x = -3$ $\therefore x = 5$	K1 - Properties K1 - Quadratic equation K1 - Factorize J1 J1

NO	ANSWER SCHEME	MARKS
2(b)	$x^2 + 10x + r \leq 0$ $(x + 5)^2 - 5^2 + r \leq 0$ $(x + 5)^2 \leq 25 - r$ $\sqrt{(x + 5)^2} \leq \sqrt{25 - r}$ $ x + 5 \leq \sqrt{25 - r}$ $x^2 + 10x + 9 \leq 0$ Compare with $x^2 + 10x + r \leq 0$, thus $r = 9$ $ x + 5 \leq \sqrt{25 - 9}$ $ x + 5 \leq \sqrt{16}$ $ x + 5 \leq 4$ $-4 \leq x + 5 \leq 4$ $-9 \leq x \leq -1$ $\therefore x \in [-9, -1]$	K1 - CTS K1 - Surd J1 K1 K1 K1 J1
TOTAL		12 marks
3(a)	$f[f(x)] = f\left[\frac{2x}{x-1}\right]$ $= \frac{2\left(\frac{2x}{x-1}\right)}{\left(\frac{2x}{x-1}\right) - 1}$ $= \frac{\frac{4x}{x-1}}{\frac{2x - (x-1)}{x-1}}$ $= \frac{\frac{4x}{x-1}}{\frac{x+1}{x-1}}$ $= \frac{4x}{x+1}$ $x + 1 = 0$ $x = -1$ $f \circ f$ is undefined when $x = -1$	K1 - Substitute J1 - Answer J1 - Find x

NO	ANSWER SCHEME	MARKS
3(b)	$f[f^{-1}(x)] = x$ $\frac{2f^{-1}(x)}{f^{-1}(x) - 1} = x$ $2f^{-1}(x) = x(f^{-1}(x) - 1)$ $2f^{-1}(x) = xf^{-1}(x) - x$ $2f^{-1}(x) - xf^{-1}(x) = -x$ $[f^{-1}(x)](2 - x) = -x$ $f^{-1}(x) = \frac{-x}{2 - x}$ $\therefore D_{f^{-1}} = (-\infty, 2) \cup (2, \infty)$	K1 - Substitute K1 - Attempt to solve J1 - Answer B1
3(c)	$g\left(\frac{2x}{x-1}\right) = \frac{8x^2 - 4x}{(x-1)^2}$ Let $u = \frac{2x}{x-1}$, $u(x-1) = 2x$ $ux - u = 2x$ $ux - 2x = u$ $x(u-2) = u$ $x = \frac{u}{u-2}$ $g(u) = \frac{8\left(\frac{u}{u-2}\right)^2 - 4\left(\frac{u}{u-2}\right)}{\left(\frac{u}{u-2} - 1\right)^2}$ $= \frac{8\left[\frac{u^2}{(u-2)^2}\right] - 4\left(\frac{u}{u-2}\right)}{\left(\frac{u-u+2}{u-2}\right)^2}$ $= \frac{8\left[\frac{u^2}{(u-2)^2}\right] - 4\left(\frac{u}{u-2}\right)}{\frac{4}{(u-2)^2}}$ $= 2u^2 - u(u-2)$ $= u^2 + 2u$ $\therefore g(x) = x^2 + 2x$	K1 - Substitute K1 - Find x K1 - Substitute x K1 - Attempt to solve J1 J1
TOTAL		13 marks

NO	ANSWER SCHEME	MARKS
4(a)	$D_f = (-\infty, \infty)$ $R_f = (1, \infty)$	B1 B1
4(b)	Let $g(x_1) = g(x_2)$ $1 + 2^{x_1-2} = 1 + 2^{x_2-2}$ $2^{x_1-2} = 2^{x_2-2}$ $x_1 - 2 = x_2 - 2$ $x_1 = x_2$ Since $x_1 = x_2$, therefore $g(x)$ is one-to-one function. $gg^{-1}(x) = x$ $1 + 2^{g^{-1}(x)-2} = x$ $2^{g^{-1}(x)-2} = x - 1$ $[g^{-1}(x) - 2] \ln 2 = \ln(x - 1)$ $g^{-1}(x) = \frac{\ln(x - 1)}{\ln 2} + 2$	K1 J1 K1 - Substitute K1 - Attempt to solve J1 - Answer
4(c)		R1 - Correct shape R1 - Correct labels R1 - All correct
TOTAL		10 marks
5(a)	$g(2) = \lim_{x \rightarrow 2^+} \sqrt{x^2 - 4} = \lim_{x \rightarrow 2^+} (kx^2 - 12)$ $4k - 12 = \lim_{x \rightarrow 2^+} \sqrt{x^2 - 4} = \lim_{x \rightarrow 2^+} (kx^2 - 12)$ $4k - 12 = \sqrt{2^2 - 4}$ $4k - 12 = 0$ $k = 3$	B1 K1 J1

NO	ANSWER SCHEME	MARKS
5(b)	$\lim_{x \rightarrow 2^+} \frac{x+2}{\sqrt{x^2-4}} = +\infty$ $\lim_{x \rightarrow -2^-} \frac{x+2}{\sqrt{x^2-4}} = -\infty$ Vertical asymptotes are $x = 2, x = -2$. $\begin{aligned} \lim_{x \rightarrow +\infty} h(x) &= \lim_{x \rightarrow +\infty} \frac{x+2}{\sqrt{x^2-4}} \\ &= \lim_{x \rightarrow +\infty} \frac{\frac{x+2}{x}}{\frac{\sqrt{x^2-4}}{x}} \\ &= \lim_{x \rightarrow +\infty} \frac{1 + \frac{2}{x}}{\sqrt{\frac{x^2}{x^2} - \frac{4}{x^2}}} \\ &= \lim_{x \rightarrow +\infty} \frac{1 + \frac{2}{x}}{\sqrt{1 - \frac{4}{x^2}}} \\ &= \frac{1+0}{\sqrt{1-0}} = 1 \end{aligned}$ $\begin{aligned} \lim_{x \rightarrow -\infty} h(x) &= \lim_{x \rightarrow -\infty} \frac{x+2}{\sqrt{x^2-4}} \\ &= \lim_{x \rightarrow -\infty} \frac{\frac{x+2}{x}}{\frac{\sqrt{x^2-4}}{x}} \\ &= \lim_{x \rightarrow -\infty} \frac{-1 - \frac{2}{x}}{\sqrt{\frac{x^2}{x^2} - \frac{4}{x^2}}} \\ &= \lim_{x \rightarrow -\infty} \frac{-1 - \frac{2}{x}}{\sqrt{1 - \frac{4}{x^2}}} \\ &= \frac{-1-0}{\sqrt{1-0}} = -1 \end{aligned}$ Horizontal asymptotes are $y = 1, y = -1$.	K1 – Find vertical asymptote J1 K1 K1 J1 - both J1
TOTAL		9 marks

NO	ANSWER SCHEME	MARKS
6(a)	$\frac{dy}{dx} = e^{-x}(-2\sin 2x) + (\cos 2x)(-e^{-x})$ $\frac{dy}{dx} = -2e^{-x}\sin 2x - y$ $\frac{d^2y}{dx^2} = -4e^{-x}\cos 2x + 2e^{-x}\sin 2x - \frac{dy}{dx}$ $\frac{d^2y}{dx^2} = -4y - y - \frac{dy}{dx} - \frac{dy}{dx}$ $\frac{d^2y}{dx^2} = -5y - 2\frac{dy}{dx}$ $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 0$	K1 - differentiate K1 – 2nd derivative K1 - substitute J1
6(b) (i)	$\frac{dx}{dt} = \frac{(t+1)(1) - t(1)}{(t+1)^2} = \frac{1}{(t+1)^2}, \frac{dy}{dt} = 1$ $\frac{dy}{dx} = 1(t+1)^2 = (t+1)^2$ $\frac{dy}{dx} = 4$ $(t+1)^2 = 4$ $t = \pm 2 - 1$ $t = -3, t = 1$ $t = 1$	K1 - Both correct $\frac{dy}{dt}, \frac{dx}{dt}$ K1J1 K1 J1
6(b) (ii)	When $t = 1$, $x = \frac{1}{1+1} = \frac{1}{2}$, $y = 1 - 1 = 0$ $m_T = 4$, $y - 0 = 4\left(x - \frac{1}{2}\right)$ $y = 4x - 2$	K1 K1 K1 J1
TOTAL		13 marks
7(a)	$v = \pi r^2 h$ $16 = \pi x^2 y$ $y = \frac{16}{\pi x^2} \quad (\text{shown})$	K1 J1

NO	ANSWER SCHEME	MARKS
7(b)	Let S_1 be the surface area of the container, $S_1 = \pi r^2 + 2\pi r h$ $= \pi x^2 + 2\pi x y$ $= \pi x^2 + 2\pi x \left(\frac{16}{\pi x^2} \right)$ $= \pi x^2 + \frac{32}{x}$	K1 – surface area J1
7(c)	Let S_2 be the surface area of the conical cover, $S_2 = \pi r \ell$ $= \pi x (2x)$ $= 2\pi x^2$ Let C be the total cost of the silo, $C = pS_1 + \frac{3}{2}pS_2$ $= p \left(\pi x^2 + \frac{32}{x} \right) + \frac{3}{2}p (2\pi x^2)$ $= \pi p x^2 + \frac{32}{x} p + 3\pi p x^2$ $= 4\pi p x^2 + \frac{32}{x} p$ $= p \left(4\pi x^2 + \frac{32}{x} \right)$ $\frac{dC}{dx} = p \left(8\pi x - \frac{32}{x^2} \right)$ $\frac{dC}{dx} = 0$ $p \left(8\pi x - \frac{32}{x^2} \right) = 0$ $8\pi x = \frac{32}{x^2}$ $x^3 = \frac{4}{\pi}$ $x = \sqrt[3]{\frac{4}{\pi}}$	K1 – surface area J1 K1 – total cost K1 – differentiate K1 – equate with zero J1

NO	ANSWER SCHEME	MARKS
7	$\frac{d^2C}{dx^2} = p \left(8\pi + \frac{64}{x^3} \right)$ When $x = \sqrt[3]{\frac{4}{\pi}}$, $\frac{d^2C}{dx^2} = p \left[8\pi + \frac{64}{\left(\sqrt[3]{\frac{4}{\pi}} \right)^3} \right] = 24\pi p > 0$ $\therefore$ C is minimum when $x = \sqrt[3]{\frac{4}{\pi}}$.	K1 – 2nd derivative K1 - substitute J1
TOTAL		13 marks